- IMPEDANCE CONDITIONS

$$\begin{cases} \alpha \dfrac{\partial u}{\partial t}(0,t) - c\dfrac{\partial u}{\partial x}(0,t) = g_0^I(t) & \leftarrow 1^{ST} \\ \alpha \dfrac{\partial u}{\partial t}(L,t) + c\dfrac{\partial u}{\partial x}(L,t) = g_L^I(t) & \leftarrow 2^{ND} \end{cases}$$

$$c = \sqrt{\frac{\mu}{\rho}} \qquad \alpha > 0$$

$\alpha = 1 \implies$ ABSORBING CONDITION

$$\int_0^L \rho\, u_{tt}\, v\, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x}\, dx$$

$$- \underbrace{\mu(L)\frac{\partial u}{\partial x}(L)\, v(L)}_{2^{ND}} + \underbrace{\mu(0)\frac{\partial u}{\partial x}(0)\, v(0)}_{1^{ST}} = \int_0^L f v\, dx$$

$$\alpha \frac{\partial u}{\partial t}(0,t) - g_0^I(t) = c(0)\frac{\partial u}{\partial x}(0,t)$$

$$\frac{\partial u}{\partial x}(0,t) = \frac{\alpha}{c}\frac{\partial u}{\partial t}(0,t) - \frac{1}{c} g_0^I(t) \quad \bullet$$

$$\frac{\partial u}{\partial x}(L,t) = \frac{1}{c} g_L^I(t) - \frac{\alpha}{c}\frac{\partial u}{\partial t}(L,t) \quad \bullet\bullet$$

$$\int_0^L \rho\, u_{tt}\, v\, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x}\, dx$$

$$- \mu(L)\left[\frac{1}{c}\, g_L^I(t) - \frac{\alpha}{c}\frac{\partial u}{\partial t}(L,t)\right] v(L)$$

$$+ \mu(0)\left[\frac{\alpha}{c}\frac{\partial u}{\partial t}(0,t) - \frac{1}{c}\, g_0^I(t)\right] v(0) = \int_0^L f\, v\, dx$$

WEAK FORM $\quad \forall t > 0$ find $u(t) \in H^1(0,L)$ s.t.

$$\begin{cases} m(u_{tt}, v) + a(u, v) + \frac{\mu}{c}\alpha \frac{\partial u}{\partial t}(L,t)\, v(L) \\ \quad + \frac{\mu}{c}\alpha \frac{\partial u}{\partial t}(0,t)\, v(0) \\ \quad = \int_0^L f\, v\, dx + \frac{\mu}{c}\, g_L^I(t)\, v(L) \\ \quad + \frac{\mu}{c}\, g_0^I(t)\, v(0) \qquad \forall v \in H^1(0,L) \\ + \text{ INIT. COND} \end{cases}$$

$$\Downarrow$$

$$M\ddot{\underline{U}}(t) + A\underline{U}(t) + S\dot{\underline{U}}(t) = \underline{F}(t)$$

$$u_h(x,t) = \sum_{j=0}^{N_h} u_j(t)\varphi_j(x)$$

$$\frac{\partial u_h}{\partial t}(x,t) = \sum_{j=0}^{N_h} \dot{u}_j(t)\varphi_j(x)$$

$\forall i,j$

$$\frac{\partial u_h}{\partial t}(0,t)\,v(0) \rightsquigarrow \dot{u}_j(t)\underbrace{\varphi_j(0)\varphi_i(0)}_{\neq 0 \text{ if only if } j=i=0}$$

$\forall i,j$

$$\frac{\partial u_h}{\partial t}(L,t)\,v(L) \rightsquigarrow \dot{u}_j(t)\underbrace{\varphi_j(L)\varphi_i(L)}_{\neq 0 \text{ if and only if } j=i=N_h}$$

$$S = \begin{pmatrix} * & & 0 \\ & \ddots & \\ 0 & & * \end{pmatrix} \qquad * = \frac{\mu}{c}\alpha$$

$$\underline{F} = \begin{pmatrix} \int_0^L f\varphi_i\,dx \\ \vdots \\ \int_0^L f\varphi_{N_h}\,dx \end{pmatrix} + \begin{pmatrix} \frac{\mu}{c} g_0^I(t) \\ 0 \\ \vdots \\ 0 \\ \frac{\mu}{c} g_L^I(t) \end{pmatrix}$$

$$S\dot{\underline{U}} \approx S\underbrace{\frac{U_{k+1}-U_{k-1}}{2\Delta T}}_{2^{ND}\ \text{ACCURATE}}$$

$$U_k = U(t_k)$$

$$t_k = k\Delta T$$

$$\left|\dot{U}(t_k) - \frac{U_{k+1}-U_{k-1}}{2\Delta T}\right| \simeq O(\Delta T^2)$$

$k \geq 2$

$$M\frac{U_{k+1}-2U_k+U_{k-1}}{\cancel{\Delta T^2}} + \overset{\Delta T^2}{A}U_k + \overset{\Delta T}{S}\frac{U_{k+1}-U_{k-1}}{2\cancel{\Delta T}} = \overset{\Delta T^2}{F_k}$$

$$\left(M + \frac{\Delta T}{2}S\right)U_{k+1} = \Delta T^2 F_k + \left(2M - \Delta T^2 A\right)U_k + \left(\frac{\Delta T}{2}S - M\right)U_{k-1}$$

$k = 1$

$$M\ddot{U}(t_0) + A\underbrace{U(t_0)}_{\hat{U}_0} + S\underbrace{\dot{U}(t_0)}_{\hat{U}_1} = F_0$$

$$M\frac{2}{\cancel{\Delta T^2}}\left(\underline{U}_1 - \hat{\underline{U}}_0 - \hat{U}_1\Delta T\right) + \overset{\frac{\Delta T^2}{2}}{A}\hat{U}_0 + \overset{\frac{\Delta T^2}{2}}{S}\hat{U}_1 = \overset{\frac{\Delta T^2}{2}}{F_0}$$

$k=1$

$$M\underline{U}_1 = \frac{\Delta T^2}{2}\left(\underline{F}_0 - A\hat{U}_0 - S\hat{U}_1\right) + M\hat{U}_0 + \Delta T M\hat{U}_1$$

$$u(x,t) = \cos(kx - \omega t)$$

$$\begin{cases} \alpha \dfrac{\partial u}{\partial t}(0,t) - c \dfrac{\partial u}{\partial x}(0,t) = g_0^I(t) \\ \alpha \dfrac{\partial u}{\partial t}(L,t) + c \dfrac{\partial u}{\partial x}(L,t) = g_L^I(t) \end{cases}$$

$$\frac{\partial u}{\partial t} = +\omega \sin(kx - \omega t)$$

$$\frac{\partial u}{\partial x} = -k \sin(kx - \omega t)$$

$\alpha = 1$ ....
TRANSP.
$\alpha > 1$

$$g_0^I(t) = \alpha \omega \sin(-\omega t) + ck \sin(-\omega t)$$
$$= -(\alpha\omega + ck)\sin(\omega t)$$

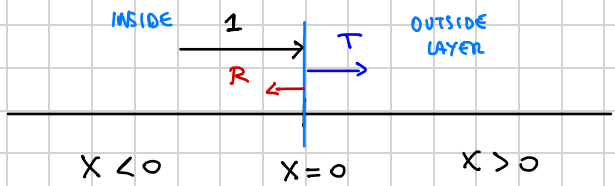
$$g_L^I(t) = \alpha \omega \sin(kL - \omega t) - ck \sin(kL - \omega t)$$
$$= (\alpha\omega - ck)\sin(kL - \omega t)$$

PML CONDITIONS

GOAL!

$T = 1$

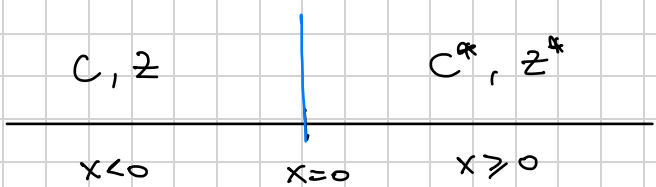
$R = 0$



$$u(x,t) = \begin{cases} e^{i(\omega t - kx)} + R\, e^{i(\omega t + kx)} & x < 0 \\ T\, e^{i(\omega t - kx)} & x \geq 0 \end{cases}$$

$$\rho(x) \frac{\partial^2 u}{\partial t^2}(x,t) - \frac{\partial}{\partial x}\left(\mu(x) \frac{\partial u}{\partial x}(x,t)\right) = 0 \qquad (W)$$

$$c(x) = \sqrt{\frac{\mu(x)}{\rho(x)}} \qquad z(x) = \sqrt{\rho(x)\,\mu(x)}$$ IMPEDANCE



IN $x = 0$
I WANT THAT

$$\begin{cases} u(0^-,t) = u(0^+,t) \\ \mu^- \dfrac{\partial u}{\partial x}(0^-,t) = \mu^+ \dfrac{\partial u}{\partial x}(0^+,t) \end{cases}$$

AFTER SOME CALCULATION...

$\Rightarrow$ REFLECTION

$$R = \frac{Z - Z^*}{Z + Z^*}$$

TRANSMISSION

$$T = \frac{2Z}{Z + Z^*}$$

$$R = 0 \iff Z = Z^*$$ IMPEDANCE MATCHING

$$T = \frac{2Z}{2Z} = 1$$ PERFECT TRANSMISSION !

How to obtain this ?

Suppose that $u(x,t) = \hat{u}(x)\, e^{i\omega t}$

$$\rho \underbrace{(i\omega)^2}_{-\omega^2} \hat{u}(x) \cancel{e^{i\omega t}} - \frac{\partial}{\partial x}\left(\mu(x) \frac{\partial \hat{u}}{\partial x}\right) \cancel{e^{i\omega t}} = 0$$

$$-\rho\omega^2 \hat{u}(x) - \frac{\partial}{\partial x}\left(\mu(x) \frac{\partial \hat{u}}{\partial x}\right) = 0 \qquad (H)$$ HELMHOLTZ EQ.

$\hat{u}(x)$ is a solution to (H) iff $u(x,t)$ is a solution for (W)

For (H) I want to chose ρ and μ to have impedance matching!

$$\begin{cases} \rho(x) & \mu(x) & x<0 \\ \rho = \dfrac{\rho(x)}{d(\omega)} & \mu = \mu(x)\, d(\omega) & x \geq 0 \end{cases}$$

$$Z(x) = \sqrt{\rho(x)\mu(x)} \rightsquigarrow Z(0^-) = \sqrt{\rho(0^-)\mu(0^-)}$$

$$Z^*(x) = \sqrt{\rho(x)\mu(x)} \rightsquigarrow Z^*(0^+) = Z(0^-)$$

$d(\omega) \in \mathbb{C}$ s.t. I have the impedance matching & reduction of amplitude for $x \geq 0$

$$\begin{cases} -\rho\omega^2 \hat{u}(x) - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial \hat{u}}{\partial x}\right) = 0 & x<0 \\ -\dfrac{\rho}{d(\omega)}\omega^2 \hat{u}(x) - \dfrac{\partial}{\partial x}\left(\mu\, d(\omega) \dfrac{\partial \hat{u}}{\partial x}\right) = 0 & x \geq 0 \end{cases}$$

- $d(\omega) = (a + ib)^{-1} \qquad a, b \in \mathbb{R}$

$$\Rightarrow \quad u(x,t) = \underbrace{e^{i\omega\left(t - \frac{ax}{c}\right)}}_{\text{PROPAGATING WAVE}} \; \underbrace{e^{\frac{\omega b x}{c}}}_{\text{DAMPING}}$$

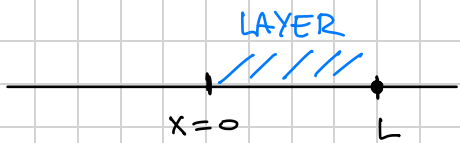
$\omega b < 0$

- $a = 1 \qquad b = -\dfrac{\sigma}{\omega} \qquad \sigma > 0 \qquad \omega b = -\sigma < 0$

$$u(x,t) = e^{i\omega\left(t - \frac{x}{c}\right)} \; e^{-\sigma \frac{x}{c}}$$

$x = 0 \qquad e^{-\frac{\sigma x}{c}} = 1$

$x \to L \qquad < 1$



$$u(L,t) = e^{i\omega\left(t - \frac{L}{c}\right)} \; \underbrace{e^{-\frac{\sigma L}{c}}}_{\ll 1}$$

If $\quad L \ll 1 \qquad \sigma \gg 1$

$\quad\;\; L \gg 1 \qquad \sigma \ll 1$

- $d(\omega) = \dfrac{i\omega}{i\omega + \sigma} \quad \Rightarrow$ CONSTANT DAMPING

$$\rho(i\omega + \sigma)\,\hat{u} - \frac{\partial}{\partial x}\left(\mu\,(i\omega + \sigma)^{-1}\,\frac{\partial \hat{u}}{\partial x}\right) = 0$$

SPACE FREQ

$x \geq 0$

SPACE
TIME

$$\rho \frac{\partial^2 u}{\partial t^2} + \underbrace{2\sigma \frac{\partial u}{\partial t} + \sigma^2 u}_{\text{ADDITIONAL TIME DOMAIN}} - \frac{\partial}{\partial x}\left(\mu \frac{\partial u}{\partial x}\right) = 0$$

$x \geq 0$

HELMHOLTZ EQ.

$$u(x,t) = \underbrace{e^{i\omega t}}\, \hat{u}(x)$$

$f(x,t) = e^{i\omega t} \hat{f}(x)$, ρ, μ are constant

$$\frac{\rho\omega^2}{\rho\omega^2} \hat{u}(x) + \frac{\mu}{\rho\omega^2} \hat{u}''(x) = \frac{\hat{f}(x)}{\rho\omega^2}$$

$$\frac{\mu}{\rho} \frac{1}{\omega^2} = \frac{c^2}{\omega^2} = \frac{1}{k^2}$$

c^2

$$\begin{cases} \hat{u}(x) + \dfrac{1}{k^2} \hat{u}''(x) = \hat{g}(x) & x \in (0,L) \\ \hat{u}(0) = \hat{u}(L) = 0 \end{cases}$$

HOW TO SOLVE IT ?

$$k^{-2}\lambda^2 + 1 = 0$$

$$\lambda^2 + k^2 = 0$$

$$\lambda_{1,2} = \pm ik$$

- $\hat{g}(x) \equiv 0$ $\quad k^{-2} \hat{u}''(x) + \hat{u}(x) = 0$

$$\hat{u}(x) = C_1 e^{ikx} + C_2 e^{-ikx}$$

$$\begin{cases} \hat{u}(0) = 0 = C_1 + C_2 \\ \hat{u}(L) = 0 = C_1 e^{ikL} + C_2 e^{-ikL} \end{cases}$$

- $C_1 = -C_2$

$$-C_2 e^{ikL} + C_2 e^{-ikL} = 0$$

- $C_2 \underbrace{\left(e^{-ikL} - e^{ikL} \right)}_{=0} = 0$

$C_2 = 0 \Rightarrow C_1 = 0$

$\hat{u}(x) = 0$ DONT WANT IT

$e^{-ikL} - e^{ikL} = 0$

$$\cancel{\cos(kL)} - \cancel{i}\sin(kL) - \cancel{\cos(kL)} - \cancel{i}\sin(kL) = 0$$

$\sin(kL) = 0$ $\qquad kL = n\pi \qquad n = 0, \dots,$

$$k_n = \frac{n\pi}{L}$$

$$\hat{u}(x) = \sum_{n=\cancel{0}\,1}^{\infty} \hat{u}_n \sin\left(\frac{n\pi}{L} x\right)$$

$$\hat{g}(x) = \sum_{n=\cancel{0}\,1}^{\infty} \hat{g}_n \sin\left(\frac{n\pi}{L} x\right)$$

$$\sum_{n=1}^{\infty} \underbrace{\hat{u}_n \left(1 - \frac{1}{k^2} \left(\frac{n\pi}{L}\right)^2 \right)} \sin\left(\frac{n\pi}{L} x\right)$$

$$= \sum_{n=1}^{\infty} \underbrace{\hat{g}_n} \sin\left(\frac{n\pi}{L}x\right)$$

$$\hat{u}_n = \frac{\hat{g}_n}{1 - \left(\frac{n\pi}{kL}\right)^2}$$

OBSERVATION

1) if $\frac{n\pi}{kL} \neq \pm 1 \Rightarrow \hat{u}_n$ is well defined

2) if $\exists\, n^* \in \mathbb{N}$ s.t. $\frac{n^*\pi}{kL} = \pm 1 \Rightarrow \hat{u}_n$ is not well defined

$g_{n^*} = 0$

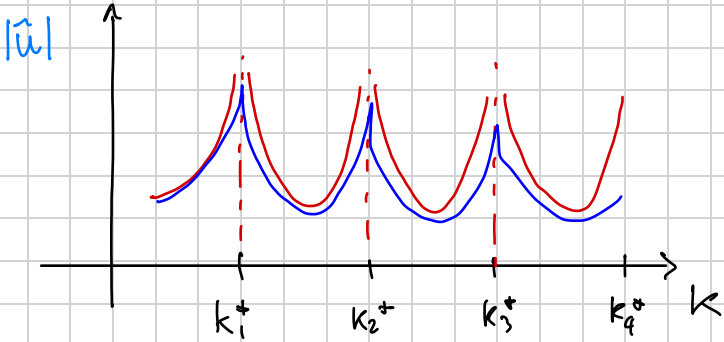
∞ SOLUTIONS

$g_{n^*} \neq 0$

NO SOLUTION

$k_n^* = \frac{\omega_{n^*}}{c}$ leads to a resonant frequency

If $c = 1 \Rightarrow k_{n^*} = \omega_n^* = \left(\frac{n^*\pi}{L}\right)^2$



K_i^* are the solutions of this problem

$$-\hat{u}''(x) = K^{*2}\,\hat{u}(x)$$

[GENERALIZED EIGENVALUE PROBLEM]

$$\rho\frac{\partial^2 u}{\partial t^2} - \mu\frac{\partial^2 u}{\partial x^2} + \gamma\frac{\partial u}{\partial t} = 0 \qquad \gamma > 0$$

$$-\rho\omega^2\,\hat{u}(x) - \mu\,\hat{u}''(x) - i\omega\gamma\,\hat{u}(x) = 0$$

$$-\hat{u}''(x) = \Big(\underbrace{\frac{\rho\omega^2}{\mu}}_{k^2} + \underbrace{\frac{i\omega\gamma}{\mu}}_{\text{IMAGINARY PART} \neq 0}\Big)\,\hat{u}(x)$$

$$-\hat{u}''(x) = \Big(k^2 + \frac{i\omega\gamma}{\mu}\Big)\,\hat{u}(x)$$

WHICH BOUNDARY CONDITIONS $x = \{0, L\}$

1) $\hat{u}(x) = g_D(x)$ SOUND SOFT CONDITION

2) $\frac{i}{\rho\omega} \hat{u}'(x) = g_N(x)$ SOUND HARD CONDITION

3) $\hat{u}(x) = Z(x) \left(\frac{i}{\rho\omega} \hat{u}'(x) \right)$

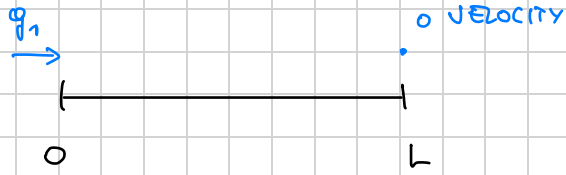
IMPEDANCE FUNCTION

$Z \equiv 1$ ABSORBING CONDIZION

$\hat{u}(x) \rightsquigarrow p(x)$ PRESSURE

$\frac{i}{\rho\omega} \hat{u}'(x) \rightsquigarrow v(x)$ VELOCITY OF PARTICLES

$$\begin{cases} \hat{u}''(x) + k^2 \hat{u}(x) = \hat{f}(x) \\ u'(0) = g_1 \\ u'(L) = 0 \end{cases}$$



$$\underbrace{\int_0^L \hat{u}''(x) v(x)\, dx}_{} + \int_0^L k^2 \hat{u}(x) v(x)\, dx = \int_0^L f v\, dx$$

$$-\int_0^L \hat{u}'(x) v'(x)\, dx + \cancel{\hat{u}'(L) v(L)}^{\,0} - \hat{u}'(0) v(0)$$

g_1

Find $u \in H^1(0,L)$ s.t.

$$-\int_0^L \hat{u}'(x) v'(x)\, dx + \int_0^L k^2 \hat{u}(x) v(x)\, dx = \int_0^L f v\, dx + g_1 v(0)$$

$$\forall v \in H^1(0,L)$$

$a(\overset{\vee}{u}, v)$ $\qquad$ $m(\overset{\vee}{u}, v)$

$$V_h = \left\{ \begin{array}{l} v \in C^0(0,L) \\ v|_{K_j} \in \mathbb{P}^1(K_j) \ \forall j \end{array} \right\}$$

$$V_h \subset H^1(0,L)$$

Find $u_h \in V_h$ s.t.

$$-a(u_h, v_h) + m(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$$

where
$$a(\hat{u}_h, v_h) = \int_0^L \hat{u}_h'(x)\, v'(x)\, dx$$

$$m(\hat{u}_h, v_h) = \int_0^L k^2 u_h v_h \, dx$$

$$F(v_h) = \int_0^L f v_h + g_1 v(0)$$

REPEAT FOR THE ALGEBRAIC FORM

$$\hat{u}_h(x) = \sum_{j=0}^{N_h} u_j \varphi_j(x) \qquad v_h = \varphi_i$$

ALGEBRAIC FORM!

$$[M - A]\, \underline{U} = \underline{F}$$

$$A_{ij} = \int_0^L \varphi_j' \varphi_i' \, dx$$

$$M_{ij} = \int_0^L k^2 \varphi_j \varphi_i \, dx$$

CONTAINS INFO ABOUT B.C.

$$\underline{F} \equiv 0 \iff MU = AU$$

$$\hat{M}_{ij} = \int_0^L \varphi_j \varphi_i \, dx \qquad \underbrace{k^2 \hat{M} U = AU}_{\text{DISCRETE EIGENVALUE PB}} \quad (*)$$

Computing λ eigenvalues for $(*)$ gives us the resonant modes